

CSE 423 Computer Graphics

Quiz -4 (20 marks)

Duration: 20 minutes

Name:	ID:	Section: 10
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1. A 3D scene contains a sphere of radius 5 centered at the origin (0,0,0). Light illuminates a surface point P = (0, 3, 4) of the sphere from a point light source of intensity (1.00, 0.90, 0.70) located at (0, 7, 7). Also, a uniform ambient illumination is present with intensity (0.20, 0.20, 0.30) throughout the whole environment. The sphere surface is viewed from position (0, 3, 9) and its material parameters are specified by ambient coefficients (0.50, 0.10, 0.10), diffuse coefficients (0.80, 0.20, 0.20), specular coefficients (0.90, 0.90, 0.90) and a shininess factor 2. Using phone lighting model-
 - a. Calculate the ambient light contribution reflected at point P. 4

$$\begin{aligned}
 I_{ambient} &= k_a \times I_a \\
 k_a &= (0.50, 0.10, 0.10) \\
 I_a &= (0.20, 0.20, 0.30) \\
 I_{ambient} &= (0.50 \times 0.20, 0.10 \times 0.20, 0.10 \times 0.30) \\
 I_{ambient} &= (0.10, 0.02, 0.03)
 \end{aligned}$$

- b. Determine **L** and **n** vectors and then calculate the diffuse light contribution at point P. 8

Surface Normal Vector ($\hat{\mathbf{n}}$)

$$\mathbf{N} = P - C = (0, 3, 4) - (0, 0, 0) = (0, 3, 4)$$

$$\hat{\mathbf{n}} = \frac{\mathbf{N}}{\|\mathbf{N}\|} = \frac{(0, 3, 4)}{\sqrt{0^2 + 3^2 + 4^2}} = (0, 0.60, 0.80)$$

Light Direction Vector ($\hat{\mathbf{L}}$)

$$\mathbf{L}_{vec} = L_{pos} - P = (0, 7, 7) - (0, 3, 4) = (0, 4, 3)$$

$$\hat{\mathbf{L}} = \frac{\mathbf{L}_{vec}}{\|\mathbf{L}_{vec}\|} = \frac{(0, 4, 3)}{\sqrt{0^2 + 4^2 + 3^2}} = (0, 0.80, 0.60)$$

Dot Product ($\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}$)

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{L}} = (0)(0) + (0.60)(0.80) + (0.80)(0.60) = 0.48 + 0.48 = 0.96$$

Diffuse Light Contribution ($I_{diffuse}$)

$$I_{diffuse} = k_d \times I_L \times \max(\hat{\mathbf{n}} \cdot \hat{\mathbf{L}}, 0)$$

$$k_d = (0.80, 0.20, 0.20)$$

$$I_L = (1.00, 0.90, 0.70)$$

$$I_{diffuse} = (0.80 \times 1.00 \times 0.96, 0.20 \times 0.90 \times 0.96, 0.20 \times 0.70 \times 0.96)$$

- c. Determine $\mathbf{R}$ and $\mathbf{V}$ vectors and then calculate the specular light contribution at point P.

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View Vector ($\hat{\mathbf{V}}$)

$$\mathbf{V}_{vec} = V_{pos} - P = (0, 3, 9) - (0, 3, 4) = (0, 0, 5)$$

$$\hat{\mathbf{V}} = \frac{\mathbf{V}_{vec}}{\|\mathbf{V}_{vec}\|} = \frac{(0, 0, 5)}{5} = (0, 0, 1)$$

Reflection Vector ($\hat{\mathbf{R}}$)

$$\hat{\mathbf{R}} = 2(\hat{\mathbf{n}} \cdot \hat{\mathbf{L}})\hat{\mathbf{n}} - \hat{\mathbf{L}}$$

$$\hat{\mathbf{R}} = (0, 0.352, 0.936)$$

Dot Product ($\hat{\mathbf{R}} \cdot \hat{\mathbf{V}}$)

$$\hat{\mathbf{R}} \cdot \hat{\mathbf{V}} = (0)(0) + (0.352)(0) + (0.936)(1) = 0.936$$

Specular Light Contribution ($I_{specular}$)

$$I_{specular} = k_s \times I_L \times (\max(\hat{\mathbf{R}} \cdot \hat{\mathbf{V}}, 0))^\alpha$$

$$\mathbf{k}_s = (0.90, 0.90, 0.90)$$

$$\mathbf{I}_L = (1.00, 0.90, 0.70)$$

$$\alpha = 2$$

$$(\hat{\mathbf{R}} \cdot \hat{\mathbf{V}})^\alpha = (0.936)^2 = 0.876096$$

$$\mathbf{I}_{specular} = (0.90 \times 1.00 \times 0.876096, \quad 0.90 \times 0.90 \times 0.876096, \quad 0.90 \times 0.70 \times 0.876096)$$